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ABOUT ME

I am currently an **Erasmus Mundus meta4.0** Master's student (Smart Manufacturing for Industry 4.0), specializing in advanced manufacturing technologies, materials engineering, and numerical modelling, after completing my Bachelor's in Mechanical Engineering from NUST, Pakistan. With substantial experience in design, fabrication, and instrumentation, I aim to solve real-world engineering challenges through analytical modelling and mechanization. I strive to leverage strong communicative and interpersonal skills to engage effectively with diverse individuals and teams at all levels.

WORK EXPERIENCE

SKY GLOBAL INTERN'24 – TURKISH AEROSPACE INDUSTRIES (TAI) – 22/07/2024 – 18/08/2024 – ANKARA, TÜRKIYE

Selected from the pool of 250+ candidates from Pakistan by TAI for Sky Global'24 Program and assigned to the Product Life Cycle Management (PLM) department of latest *5th Generation Turkish Fighter Jet* (**TF-KAAN**) as **Structural Design Intern**.

▪Performed topology optimization of mounting brackets and swing arm assembly in landing gear section on **Siemens NX,** reducing individual bracket weight by **~23%**.
▪Attended workshops on "Flight General Knowledge" and "Technology Readiness Levels", as part of vocational training.
▪Visited production lines of **ANKA UAVs**, **Gokbey** and **ATAK** Helicopters, gaining insights into advanced composites application, and precision assembly.
▪Observed the aerodynamic performance analysis of ANKA UAVs in **wind tunnel testing**.

SUMMER INTERN – FAUJI FERTILIZER COMPANY (FFC) LIMITED – 20/07/2023 – 06/09/2023 – SADIQABAD, PAKISTAN

Performed tasks on-site as Mechanical Engineering Intern in both **Equipment Urea-I** and **Machinery Ammonia-II** at **FFC Sadiqabad Plant** for *3 weeks each.*

▪Analyzed different types of *valves, heat exchangers, pipes, and fittings, pumps (Centrifugal and PD)* in the site vicinity and consulted relative **APIs**.
▪Observed different safety standards and protocols followed to ensure personal safety and equipment reliability.
▪Briefly examined the process of urea formation, supply of utilities (*CO_2, Ammonia, Air etc.*) to reactor.

SUMMER INTERN – NISHAT POWER LIMITED (NPL) – 01/07/2023 – 20/07/2023 – PATTOKI, PAKISTAN

Served as a Mechanical Maintenance Intern in the *Maintenance Department* at a **200 MW** *combined cycle powerplant (9x* **19.1 MW** *Wartsila engines and 1x* **18.7 MW** *Peter Brotherhood Steam turbine).*

▪Observed the **72k** major maintenance of **18v46** HFO powered Wartsila Engine.
▪Studied **P&I** diagrams of *boiler, cooling tower, HFO/LFO flowlines and Heat Recovery Steam Generation (HRSG) systems.*
▪Documented and presented a detailed report on improving the *efficiency of process* by **~3%.**

EDUCATION AND TRAINING

20/02/2026 – CURRENT Turin, Italy
ERASMUS MUNDUS JOINT MASTERS (META4.0) Politecnico di Torino - 2nd Semester

SGPA: 104/110

▪ Observed the processing, heat treatments, microstructural analysis, and composition based job-specification of Titanium, Copper, Nickel, and Aluminum based alloys.
▪ Gained expertise in multi-scale materials design and sustainability, leveraging **CES (Ansys Granta)** and **Thermo-Calc** to drive data-backed material selection and phase equilibria modeling.

▪ Developed deep competencies in **failure analysis,** applying root-cause investigation techniques and experimental/virtual lab diagnostics to hypthesize actual failure scenario.
▪ Studied **advanced forming** technologies, spanning **additive manufacturing**, machining, and powder metallurgy across metals, polymers, and composites.
▪ Executed applied engineering projects considering performance, cost, and **TRL-based sustainabilit**y, highlighted by the comprehensive material selection for an SEM Vacuum O-ring.

Website https://www.polito.it/ | **Level in EQF** EQF level 7

15/09/2025 – 12/02/2026 Saint Etienne, France
ERASMUS MUNDUS JOINT MASTER'S (META4.0) Ecole centrale de Lyon (ENISE) -1st Semester

SGPA: 17.15/20

▪ Developed expertise in **subtractive processes** (turning and drilling), evaluating **tool-material pair** interactions, machining control, and numerical modeling for process' results estimation.
▪ Focused on tribology and **surface integrity analysis**, correlating physical machining dynamics to final surface roughness and resultant structural properties.
▪ Explored the thermodynamics of **laser-material interactions** and high-temperature processes, bridging metallurgy and phase transformations with modern additive manufacturing techniques.
▪ Executed a **transverse industrial project** integrating both additive and subtractive manufacturing routes, culminating in comprehensive physical material characterization.

Website https://ec-lyon.fr/enise | **Level in EQF** EQF level 7

15/11/2021 – 10/06/2025 Islamabad, Pakistan
BE MECHANICAL ENGINEERING National University of Sciences and Technology, Islamabad

CGPA: 3.94/4.00 **(Presidential Gold Medalist)**

▪ Gained knowledge regarding instrumentation and mechanical design, developing expertise in softwares like **SolidWorks** *(CAD, Stress and Fatigue Analysis)*, **Ansys** *(FEA, CFD and Topology optimization)*.
▪ Developed strong understanding of *thermodynamics, fluid mechanics and heat and mass transfer principles*, implementing them in different technical projects.
▪ Improved understanding of various *manufacturing processes and material selection techniques.*
▪ Enhanced *technical writing* and *project management skills*, with an emphasis on sustainable practices and effective resource allocation

Website https://nust.edu.pk/ | **Level in EQF** EQF level 6

30/01/2023 – 02/06/2023 Islamabad
MINOR IN "INTRODUCTION TO MARKETING (MKT-201)" NBS-NUST

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C2	C1	C1	C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

PROJECTS

19/03/2024 – 10/06/2025
High-Precision Tracking Platform for SONAR

For my Final Year Project, I am working with a team of four to:
▪ Develop a precision tracking platform for SONAR capable of **1mm accuracy** over (6x10)m planar span with **1-degree** rotational accuracy. All the design and simulations are performed on **Ansys Fluent** and **SolidWorks**.
▪ Create a **LABView** based Human Machine Interface **(HMI)** control panel to control all onboard equipments (servo and stepper motors, winches), providing control to both on-board operator and (as safety protocol) control room operator.
▪ For data acquisition from SONAR and on-board encoders, using National Instruments DAQ cards with data handling through LABView code and **Microsoft Excel**.

09/09/2023 – 10/06/2025
"Offroad Vehicle" for BAJA SAE'24-Michigan and SMME Piston Cup

Being a member of **Chassis** and **Powertrain** at **Team Markhor**

▪ With a group of *four,* engaged in designing chassis for vehicle using **SolidWorks,** selecting among different materials *(4130 Steel, ASTM A53-Grade B)* through *grade matrix* and performed *FEA* on BAJA SAE and NCAP standards.
▪ Achieved weight reduction of **~4 kg** through weld improvements, optimized truss design *(adding gussets and removing extra zero force members)* to achieve overall weight of **~58.9 kg.**
▪ Utilized **SolidWorks** and **Ansys Fluent** simulation for optimal redesign of rear-mounted engine and front/rear differential housing, addressing inaccessibility and bending failure.
▪ Implemented dual shifter mechanism for engaging reverse using concepts from *Machine Design.*
▪ Performed suspension selection by developing a **MATLAB** based model with **GUI** to select optimum coil overs and damper for vehicle using concepts from *Mechanical Vibrations* Course.
▪ Analyzed vehicle performance on **Carsim** to simulate ride quality and maneuverability on different terrains and weather conditions and directed team to make necessary adjustments.

09/05/2023 – 10/06/2025
"Fixed Wing UAV" for DBFC'17 and IMechE UAS

Being a member of **Wings and Tails** at **Team Irtifah,** helped building a fixed-wing drone, capable of delivering **~2.5 kg** medical supplies over **~1.5 km radius** area of distress**.**

▪ Working with a team of *2,* helped in selection among *4 airfoils* **(MH-114, E-214, FX-63137, NACA 6412)** for wings by creating a *design matrix (C_d min, C_l max, C_m, α_s, and Stall quality),* performing analysis at different *speeds, angle of attacks* using **Ansys Fluent.**
▪ Selected **NACA-0012** for *rudder and vertical stabilizer,* calculated payload and propeller placement, and finalized overall wing design through **XFLR-5.**
▪ Optimized *AR (6-10) and S_w (60-90 g/dm^3)* using **MathCAD** model to achieve *6.3 AR and 71.3 g/dm^3 S_w.*

05/02/2024 – 20/02/2024
"First Person View (FPV) Quadcopter" for IAM3D

Working with a team of ***5***, helped in designing quadcopter using only *additive manufacturing.*

▪ Performed *Topology Optimization* on **Ansys Workbench** to achieve overall mass of **~312.6g** within requirements of *500g.*
▪ Employed **PLA based 3D printing** for drone fabrication.

06/2022 – 08/2022
"Emergency Line Launcher" for IMechE-AKDC

Designed and prototyped a Line Launcher using principles of statics, dynamics, projectile motion and *SolidWorks***.**

▪Led a team of *five* through the model development phase using modelling and simulations on **SolidWorks**, utilizing *Weldments, BOM and Surfacing*.
▪Accomplished optimal projectile range of **~3m** through numerous iterations of various positioning coordinates.

SKILLS

Development Tools

SolidWorks | AutoCad 2D -3D | Autodesk Fusion 360 | Ansys Workbench, Fluent and CFD (Computational Fluid Dynamics | CAD modelling with Catia V5 | XFLRv5 | MathCAD (Basic) | Matlab/Simulink (Basic) | SIEMENS NX (Good) | NI Labview | Carsim (basic)

Programming Languages

C++ (Basic) | Python (Basic)

Supplementary

Microsoft Office (Word , Excel and Power Point) | Canva

HONOURS AND AWARDS

09/11/2025
Presidential Gold Medal – National University of Sciences and Technology, Pakistan

Awarded the **President's Gold Medal** for scoring the highest GPa overall among the batch. (3.94/4.00)

24/03/2025
Erasmus Mundus Scholarship – EMJM meta4.0 Consortium (EU Commission)

Selected among **1200+** applicants worldwide (600 from Pakistan) for the fully funded Erasmus Mundus scholarship to study "manufacturing 4.0 by intelligent and sustainable technologies".

12/12/2024
High Achievers Award – National University of Sciences and Technology, Islamabad

8x awardee of GPA-based Merit scholarship during undergraduate.

20/07/2024
Sky Global Program'24 – Turkish Aerospace Industries (TAI)

Selected as *Summer Intern* at Turkish Aerospace Industries (TAI) in Ankara, Türkiye from a pool of **250+** candidates for *one month including Accommodation, Travelling, and Monthly Stipend*.

15/03/2024
ASME-EFX (IAM3D)

Top 10 finishers in among 20+ teams at national level. (3D printed FPV Quadcopter)

18/11/2023
AIAA-GIKI Chapter (DBFC'17)

Top 5 finishers among 30+ teams at national level. (Fixed-Wing UAV).

08/11/2023
SMME Piston Cup'23

1st Runner's Up among 15+ teams from different universities across Pakistan. (Offroad vehicle)

10/07/2023
Teknofest-Türkiye'23

Sole Qualifiers for Stage II from Pakistan (Fixed-Wing UAV).

06/06/2022
IMechE-NUST Chapter (AKDC)

National Qualifiers among 50+ teams from different universities across Pakistan. (Emergency Line Launcher)

CONFERENCES AND SEMINARS

21/10/2023 – 21/10/2023 Online (Microsoft Teams)
Mastering 5S and Kaizen, the Toyota Way

▪ Learnt about the 5S *(Sort, Set in order, Shine, Standardize and Sustain)* principles followed by Toyota Japan and **"Kaizen-*better change*"** in their production techniques.
▪ Discovered ways to implement *Kaizen* in daily life tasks.

07/12/2024 – 07/12/2024 Islamabad, Pakistan
DesignFest'24

▪ Gained insights into the power of SolidWorks in designing products related to *(healthcare, infrastructure, transport),* contributing to a sustainable future.
▪ Video call with **CEO** of SolidWorks **"Dr, Manish Kumar"**, hearing his point of view about current progress and role of younger generation in shaping the world.

SOCIAL AND POLITICAL ACTIVITIES

05/09/2022 – 30/03/2023 Islamabad
ASME-NUST Chapter (Executive-Admin Events)

Managed "*James Henry Design Challenge*" event with **100+ individuals**, working with team of ***6*** to organize event's logistics and operations

08/05/2023 – 08/06/2023 Islamabad
AKS-UET Lahore (Brand Ambassador-Photography Challenge)

Operated as a *liaison* for event details dissemination in the campus via *posters and mails.*

05/06/2023 – 20/09/2023
FICS'23 (Finding Innovative Creative Solutions): Executive-Team Protocols

Served as an MOC, managing connection between judges and 15+ teams of assigned portfolio.

VOLUNTEERING

01/01/2010 – 01/03/2010 SAHARA Trust, Lahore
Student Volunteer

Collected funds from different schools around the area for underprivileged students with the help of my teachers for "SAHARA Education Fund".

05/04/2024 – 21/06/2024 Sunbeams, Islamabad

Student Volunteer

Participated in "Zero out of School" campaign to encourage parents from underdeveloped sectors of Islamabad to send their children to schools for education.

05/03/2024 – 12/03/2024 EDHI Homes, Islamabad

Orphanage Visit

Regularly visited orphans, taught them different group games and brought snacks and toys for them.